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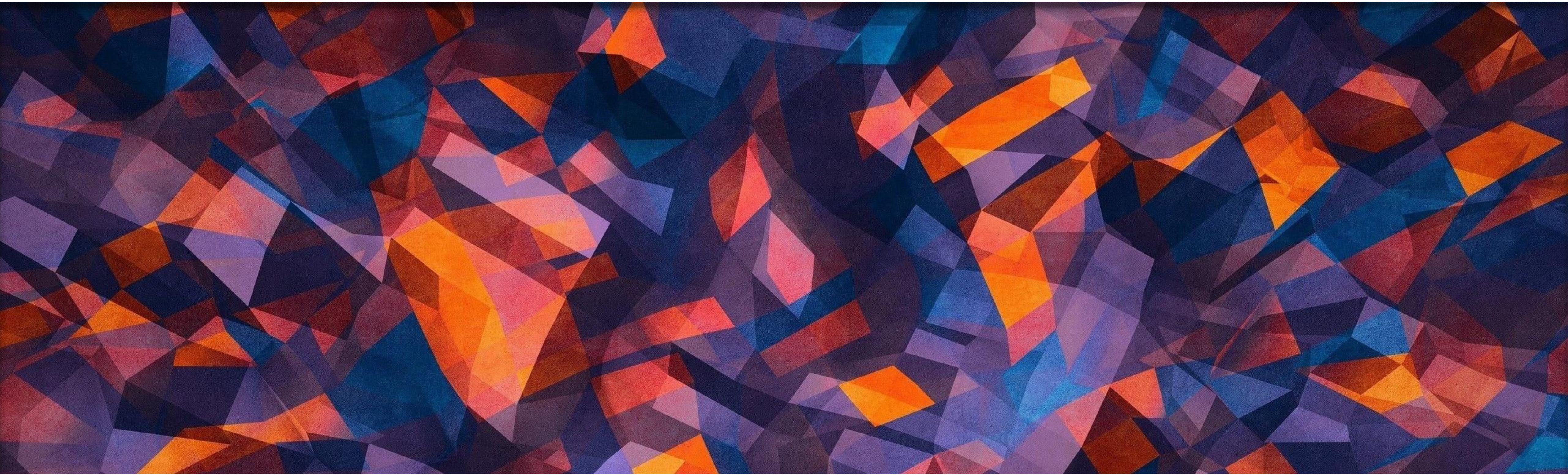
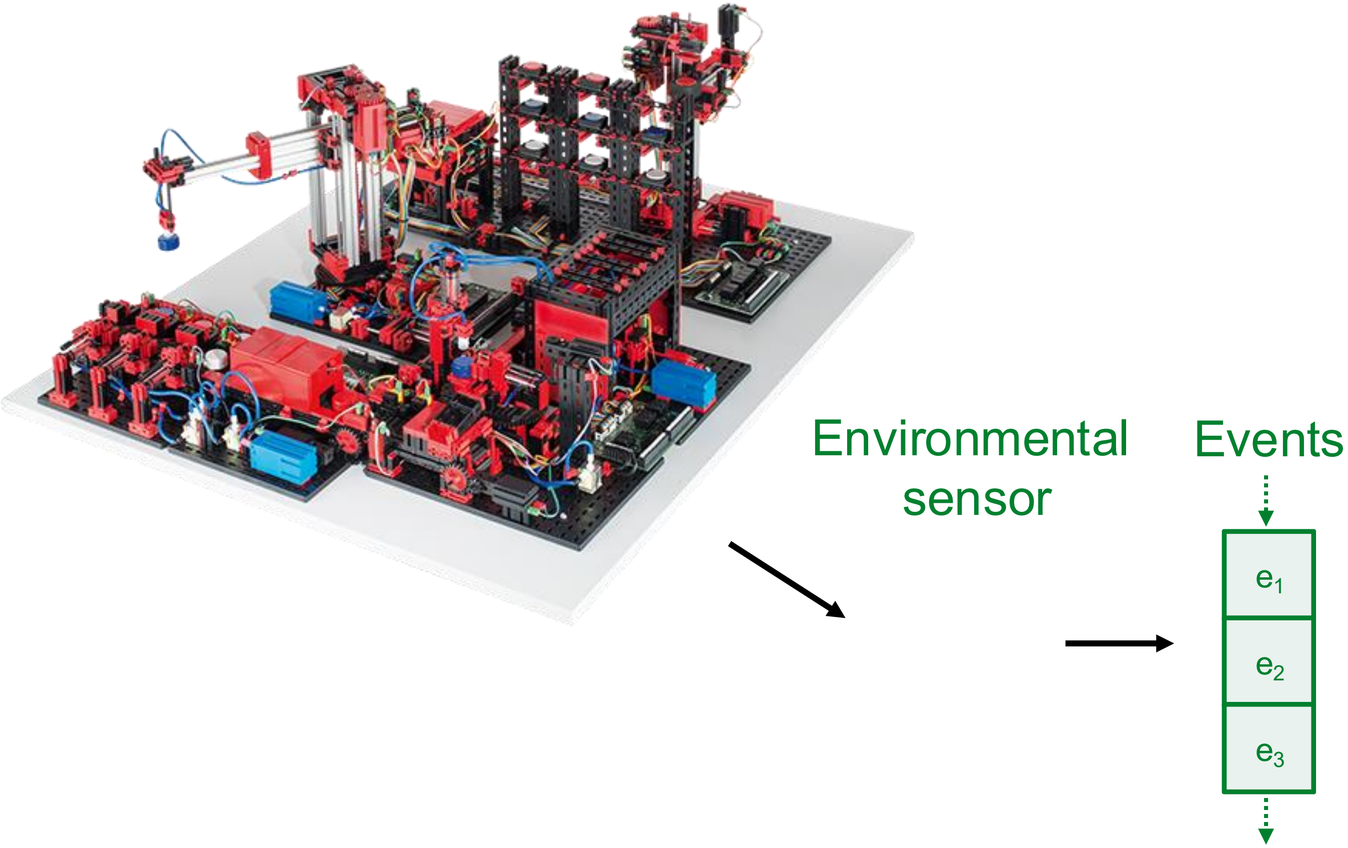


Event-driven and Process-oriented Architectures

Stream Processing – Introduction and Key Concepts

Lecture 7

Smart Factory as an Event Source



A smart factory contains many sensors that continuously produce measurements from sensors.

Each measurement represents an **event** that occurs at a specific point in time.

In this example, we focus on one environmental sensor. The events produced by this sensor form a **data stream** that can be processed as they arrive.

What is an Event?

An event is anything that we can observe occurring at a particular point in time.

Event Streams in Action. Real-time event systems with Kafka and Kinesis. Alexander Dean, Valentin Crettaz, Manning 2019

Manufacturing

Machine 21-X breaks its drill at 15:27:37 on April 15, 2019

Travelling

Patricia Smith checks in to Hotel Paradise at 19:27 December 17, 2018

Retail

User Peter77 adds glasses to shopping basket at 13:21 January 5, 2019

Pharmacy

Pharmacy GetWell opens at 08:30 January 10, 2020

Data Stream (aka Event Stream or Streaming Data)

A data stream is an an **unbounded sequence of events.**

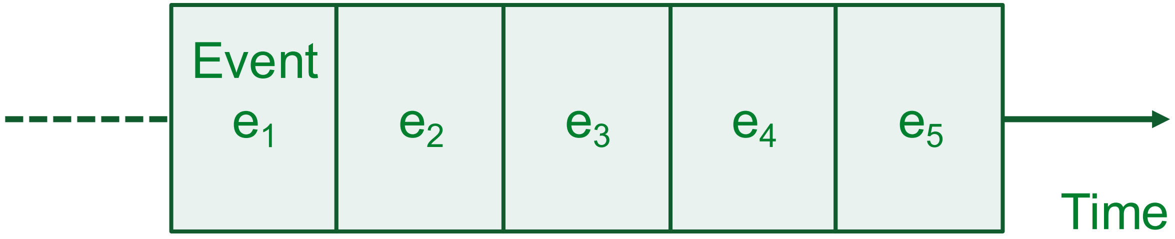
- The steam may have started before we began observing it
- The stream continues indefinitely into the future

What are the consequences of an unbounded stream for processing?

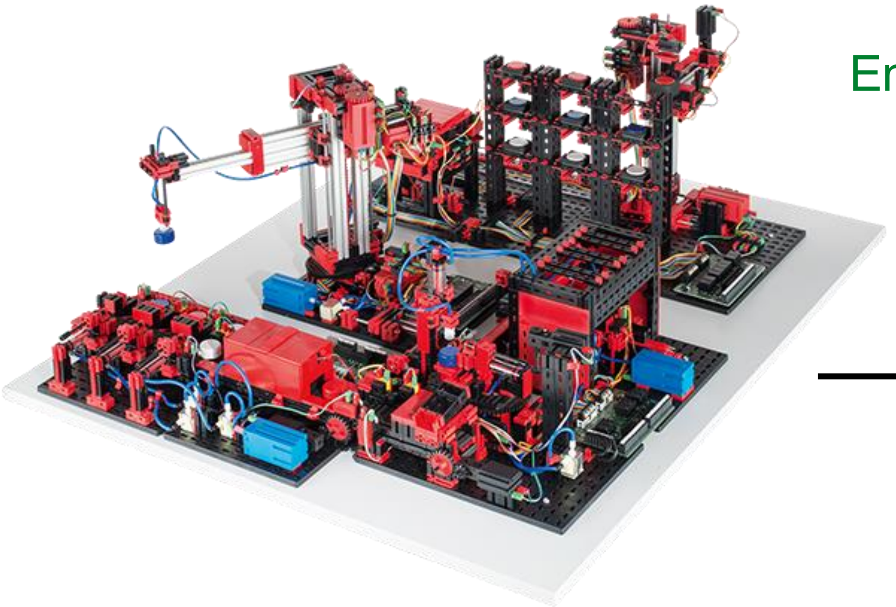
Additional Characteristics of a Data Stream

A data stream is an an **unbounded sequence of events.**

- Event streams are **ordered**
  - There is an inherent notion of which event occurred before or after other events
- **Immutable** data records
  - Events cannot be modified after they occur
- Event streams are **replayable**
  - Replayability constitutes a desirable property



Event Schema of the Environmental Sensor



Environmental sensor

Events

e_1
e_2
e_3

e_1

2020-01-21 10:44:32

t	h	p	iaq
23,10	27,90	958,50	50

e_2

2020-01-21 10:44:59

t	h	p	iaq
23,40	27,90	958,50	52

e_3

2020-01-21 10:45:17

t	h	p	iaq
23,30	27,90	958,50	52

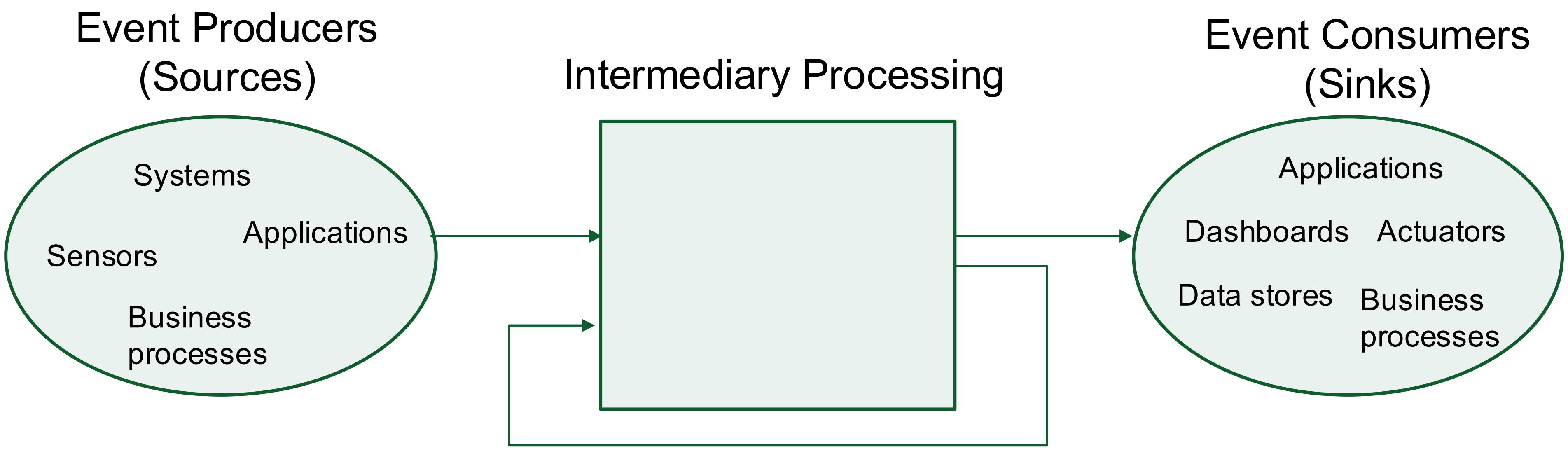
Events produced by a sensor share a common structure called an **event schema**. This schema is repeated for every event in the stream.

Each event generated by the environmental sensor contains:

- timestamp
- temperature (t)
- humidity (h)
- air pressure (p)
- air quality index (iaq)

Architecture of an Event Processing System

Event processing performs operations on events as they arrive. Typical operations include filtering, transforming, aggregating, and joining event streams.

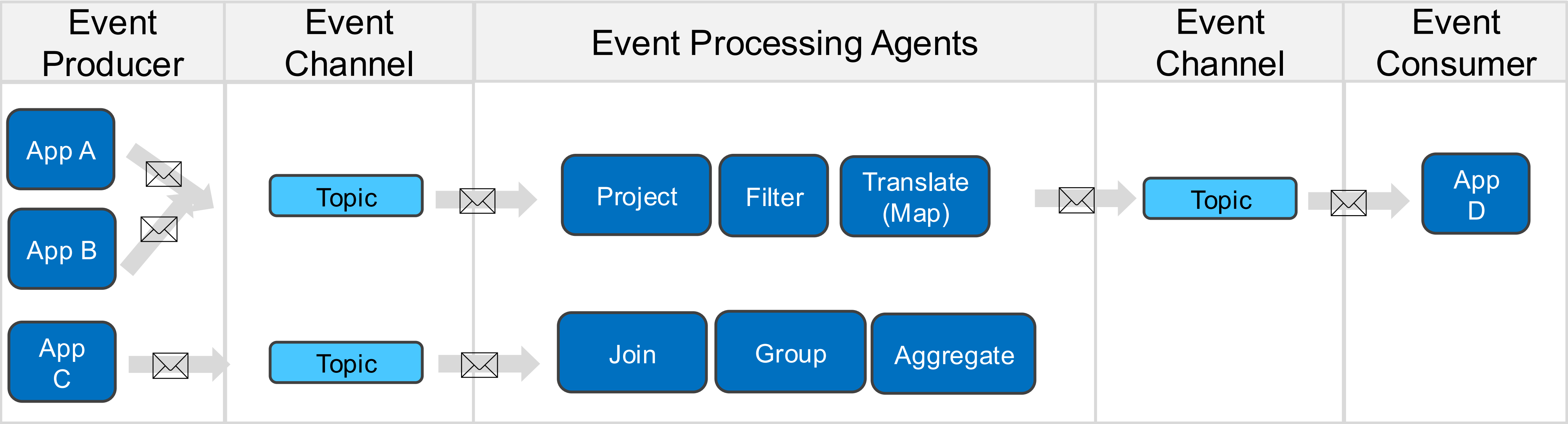


Event Processing in Action. Opher Etzion and Peter Niblett, Manning 2010.

Event Processing Networks

Event processing is typically composed of multiple processing steps. These steps together form an **event processing network.**

Event Processing Network:
A set of producers, processors, and consumers connected by event channels.



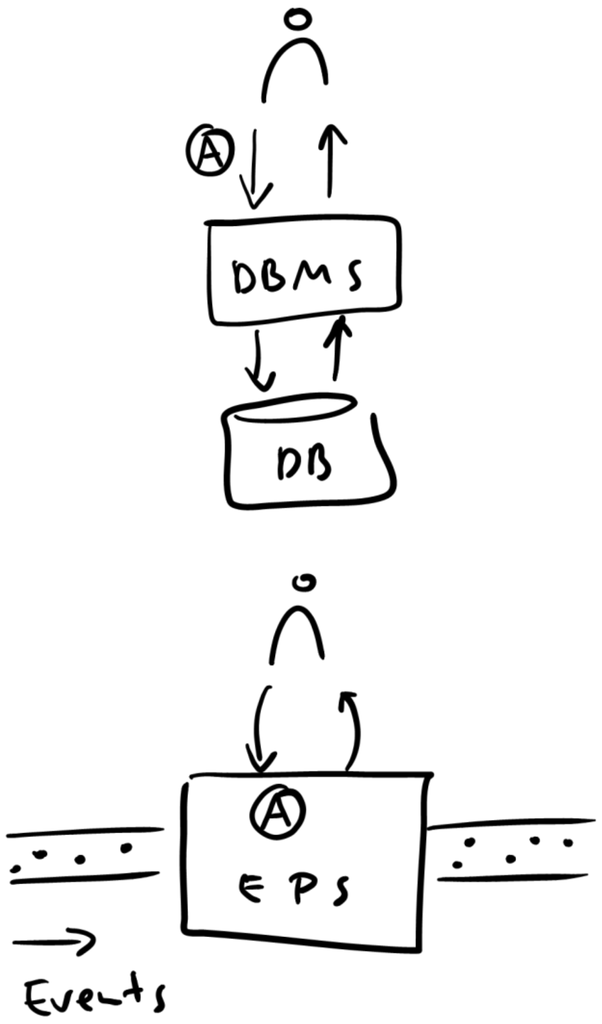
Events flow through multiple processing stages forming a pipeline.

Event Processing in Action. Opher Etzion and Peter Niblett, Manning 2010.

Request-Response vs. Batch Processing vs. Stream Processing

Request-Response	Batch Processing	Stream Processing
• Low latency (milliseconds) • Client requests → system responds • Blocking interaction **Examples**: point-of-sale systems, credit card processing	• Scheduled processing • High throughput • Results available later **Examples**: data warehouses and business intelligence reporting	• Continuous processing • Non-blocking interaction • Reacts to events in real-time **Examples**: fraud detection, monitoring, pricing

Databases vs. Stream Pricessing



- Databases
  - –Data is stored
  - –Queries run on demand
- Stream Processing
  - –Queries run continuously
  - –Results produced when events arrive

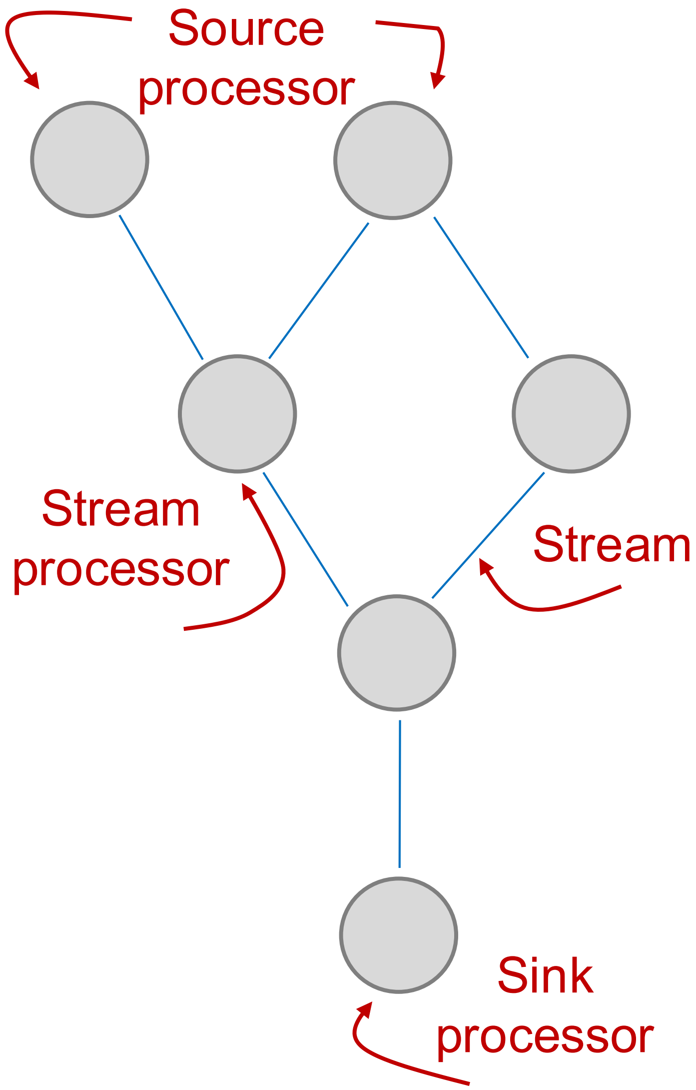
Fundamental Stream Processing Concepts

- Topology
- Time
- State
- Stream-Table Duality
- Time Windows
- Processing Gurantees

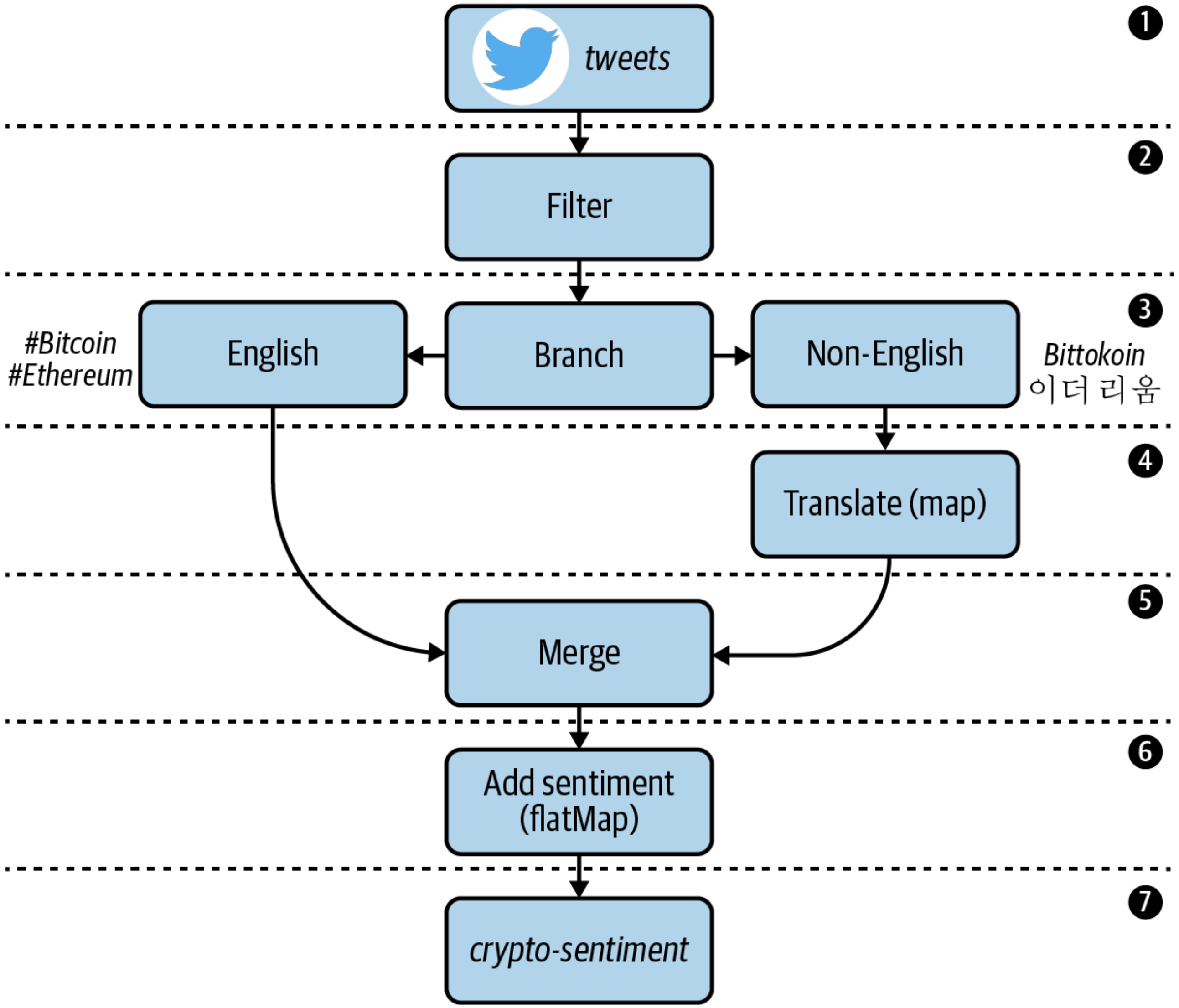
Topology

A stream processing application is organized as a **topology:**

- Source streams
- Stream processors (depicted as nodes)
- Sink streams
- Events flow through processors, forming a processing graph



Example Topology



For implementation:
https://github.com/mitch-seymour/mastering-kafka-streams-and-ksqldb

Source: Mastering Kafka Streams and ksqlDB

Time

- Time is probably the most important concept in stream processing
  - Recommended reading highlighting the complexity around time in distributed systems: „There is No Now“

- Stream applications often perform operations on time windows, thus, having a common notion of time is critical

- Different time semantics in stream processing systems: **event time, log append time** and **processing time**

Time Semantics

- **Event time**
  - Time when the event we are tracking occurred and the record was created by the data source (typically most relevant)
- **Log append time / Ingestion time**
  - The time when the event arrived at the Kafka broker, i.e., the ingestion time; use ingestion time when event time cannot be used
- **Processing time**
  - The time the stream processing application received the event to perform some calculation; can be many minutes or even hours after event time

State

Stateless Event Processing
- Each event processed independently
- No memory of previous events
- Examples: filtering, routing, transformation

Stateful Event Processing
- Processing depends on multiple events
- Requires maintaining state
- Examples: counting, joins, moving averages

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Stream-Table Duality

- Stream: sequence of changes
- Table: current state derived from changes
- Materializing a stream: converting a stream into a table

Stream-Table Duality

Representation of event stream

- “Shipment arrived with red, blue, and green shoes.”
- “Blue shoes sold.”
- “Red shoes sold.”
- “Blue shoes returned.”
- “Green shoes sold.”

Stream of changes to inventory

Shipment	Blue shoes	300
	Red shoes	300
	Green shoes	300
Sale	Blue shoes	299
	Red shoes	299
Return	Blue shoes	300
Sale	Green shoes	299

Table/materialized view representing the latest state of inventory

Red shoes	299
Blue shoes	300
Green shoes	299

Time Windows

- Most operations use time windows
  - moving averages, joins, aggregations

- Key Parameters
  - Size of the window
  - How often does the window move
  - How long does the window remain updated (grace period)

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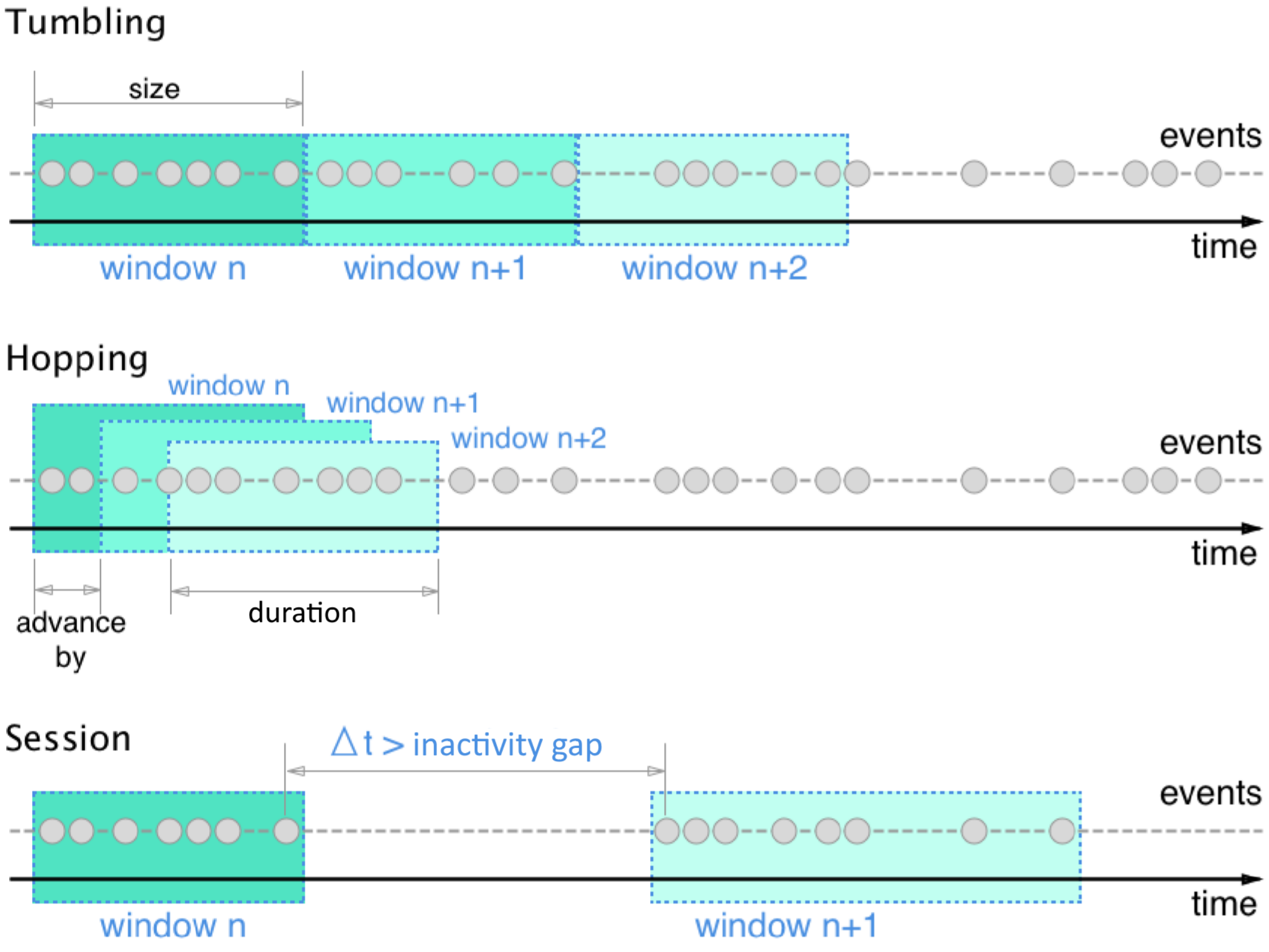
Types of Windows

- Tumbling Window
  - Time-based
  - Fixed-duration, non-overlapping, gap-less windows
- Hopping Window
  - Time-based
  - Fixed-duration, overlapping windows
- Session Window
  - Session-based
  - Dynamically-sized, non-overlapping, data-driven window



Source: https://docs.ksqldb.io/en/latest/concepts/time-and-windows-in-ksqldb-queries/

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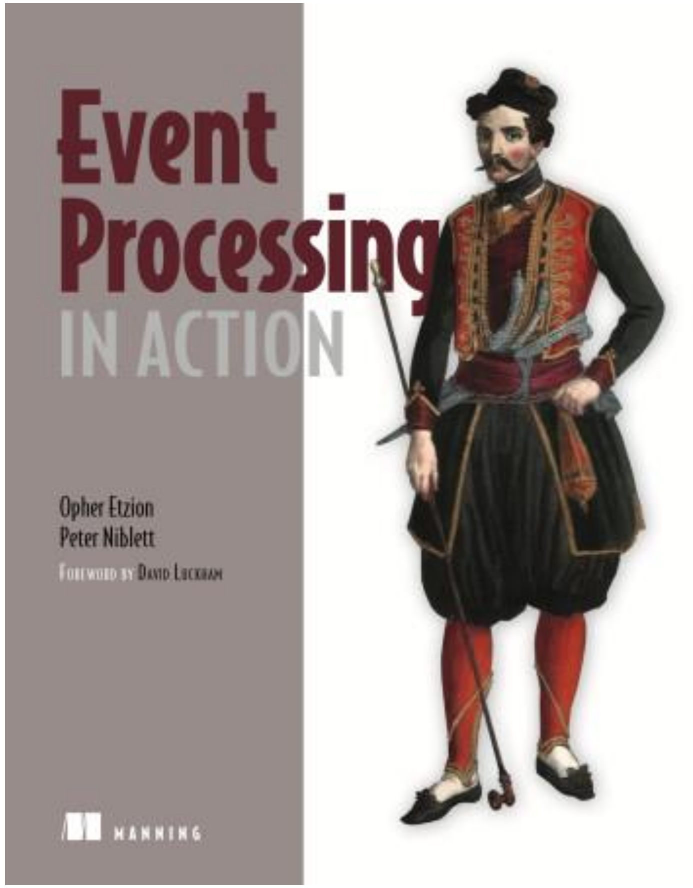
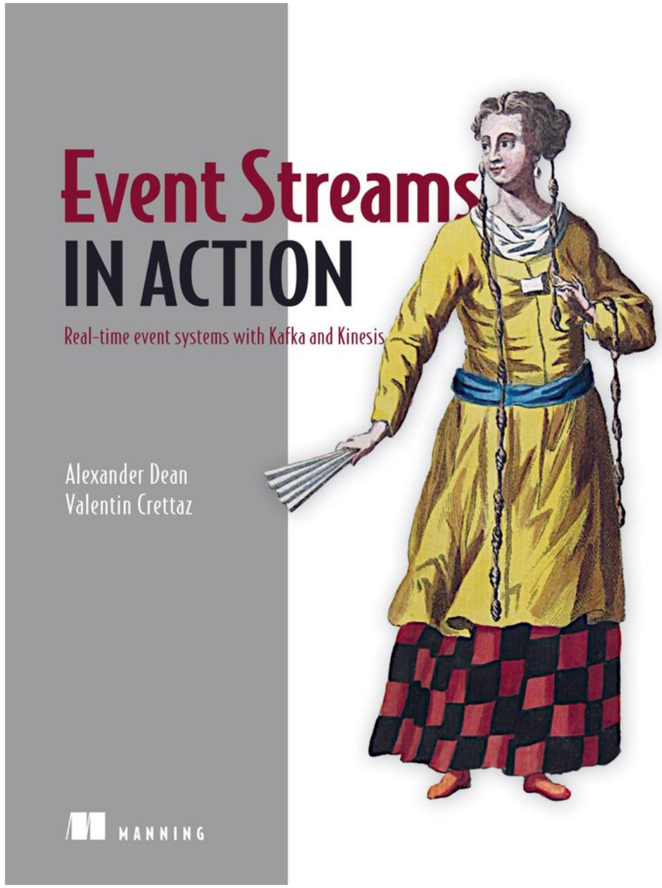
Processing Guarantees

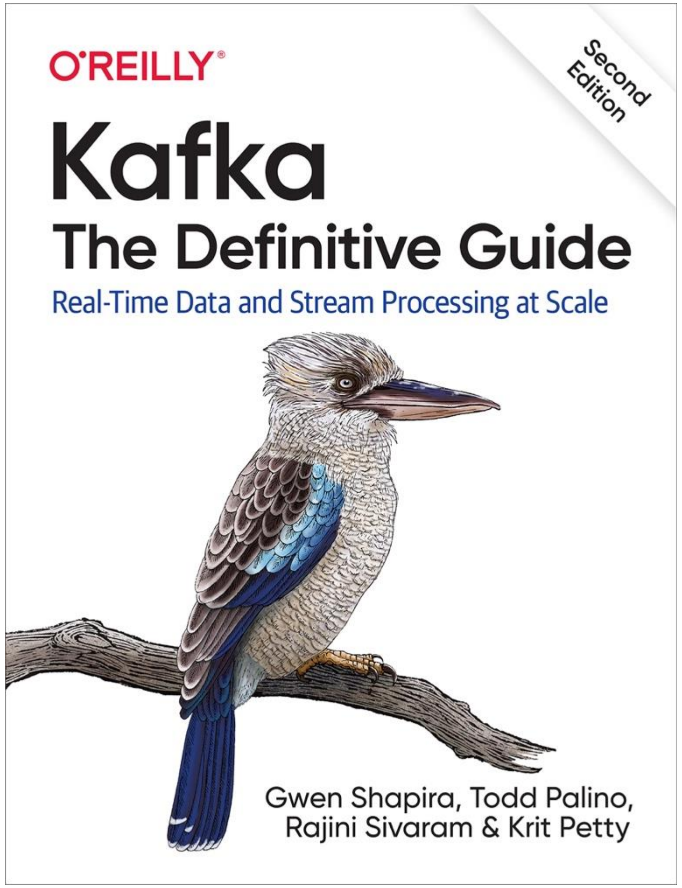
- Processing guarantees
  - At-most-once
  - At-least once
  - Exactly-once

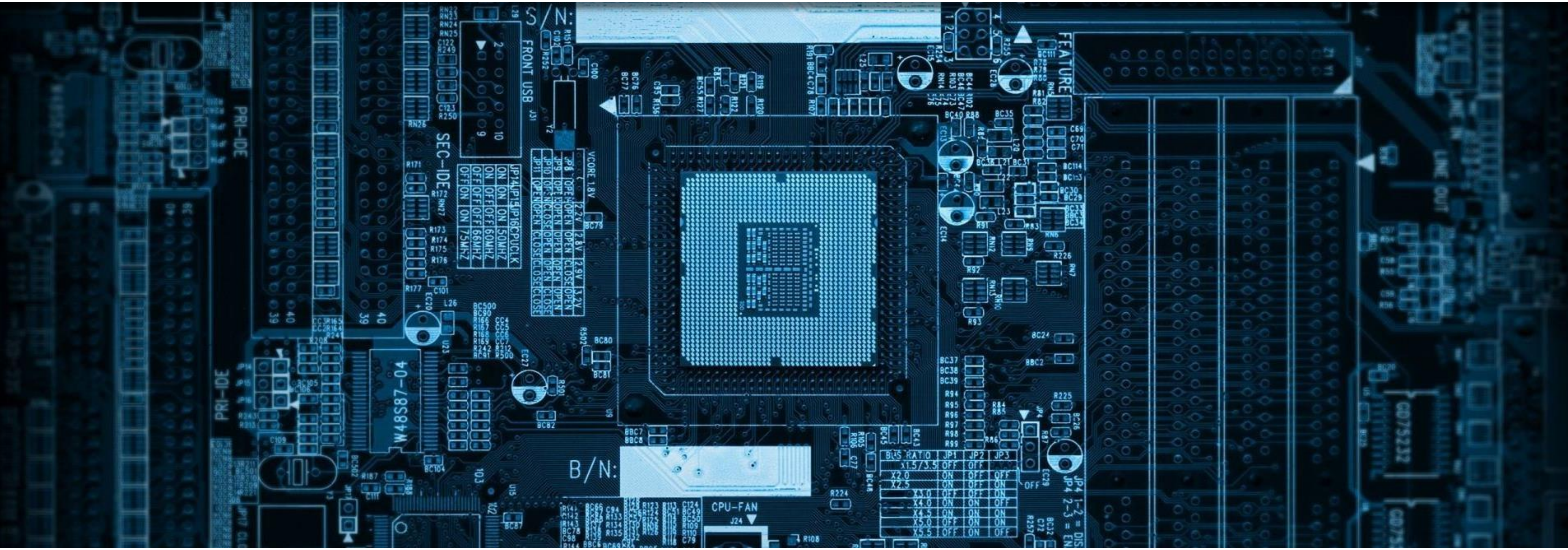
- Exactly-once ensures accurate results, but side-effects may still occur

References and Further Materials









Questions?